



Microstructure evolution and brazing mechanism of (Ti_{0.2}Zr_{0.2}Ta_{0.2}Nb_{0.2}Cr_{0.2})C high entropy ceramic joint brazed with Ni-based filler



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Abstract

High-entropy ceramics (HECs) have attracted considerable interest due to their distinctive characteristics and promising applications. Developing reliable joining techniques is essential for facilitating their widespread application. In this study, a method for brazing high-entropy ceramic (Ti_{0.2}Zr_{0.2}Ta_{0.2}Nb_{0.2}Cr_{0.2})C using nickel-based fillers was proposed, and the microstructural changes and shear strength of the brazed joints were studied in detail. The brazing seam primarily consisted of Cr₇Ni₃, Cr₂₃C₆, and Ni(s,s). The interfacial bonding was mainly influenced by the formation of a Cr₂₃C₆ reaction layer and the

diffusion of Ni into the HEC matrix. Notably, the diffusion of Ni induced a phase transformation from the original HEC to an HEC(Ni diffused) phase with a similar lattice structure. This study presented a novel approach to addressing the issues of inadequate wettability and subpar joint strength in brazing connections of high-entropy ceramics. This facilitated the formation of a semi-coherent interface with a reduced lattice mismatch at the Cr₂₃C₆/HEC interface on the high-entropy ceramic side. Additionally, the favorable plastic deformation ability of Cr₂₃C₆ enhanced joint performance. Consequently, joints brazed at 1100°C for 10min achieved a maximum shear strength of 124MPa.



Keywords

High entropy ceramic; Brazing; Microstructure evolution; Mechanical properties

1. Introduction

High entropy ceramics (HEC) have advantages such as high strength and excellent high temperature stability [[1], [2], [3]]. These attributes have generated considerable research interest, establishing HECs as promising candidates for structural materials in extreme environments [4,5]. The field gained momentum after Rost et al. first reported (Co_{0.2}Cu_{0.2}Mg_{0.2}Ni_{0.2}Zn_{0.2})O high-entropy oxide ceramics in 2015 [6]. Subsequent studies have expanded to include high-entropy borides, silicides, and carbides, all exhibiting enhanced thermomechanical properties and entropy-stabilized thermodynamic resilience [[7], [8], [9]]. Potential applications encompass drill bits, cutting tools, and structural components for aerospace and nuclear industries.

Among these, high-entropy carbide ceramics combine the excellent properties of refractory metal carbides. In comparison to single-component carbide ceramics, high-entropy carbide ceramics exhibit exceptional high-temperature strength, oxidation resistance, and radiation resistance [10,11]. Wang et al. [12] explored the mechanisms behind irradiation damage in high-entropy ceramics. The microstructure of (Zr_{0.25}Ta_{0.25}Nb_{0.25}Ti_{0.25})C, affected by Zr ion irradiation, was characterized by defect clusters measuring a few nanometers in diameter, and it exhibited high phase stability, as evidenced by the X-ray diffraction results.

Kavak et al. [13] utilized spark plasma sintering technology to fabricate (HfMoTiWZr)C high-entropy carbide ceramics, which exhibited high hardness and wear resistance. The high-entropy carbide ceramics exhibited remarkable phase stability, impressive hardness, and outstanding resistance to irradiation. However, the brittleness of high-entropy ceramics complicates the manufacturing of large and complex-shaped components [14,15]. Therefore, developing a reliable joining technique for high-entropy ceramics is of great significance for their widespread application.

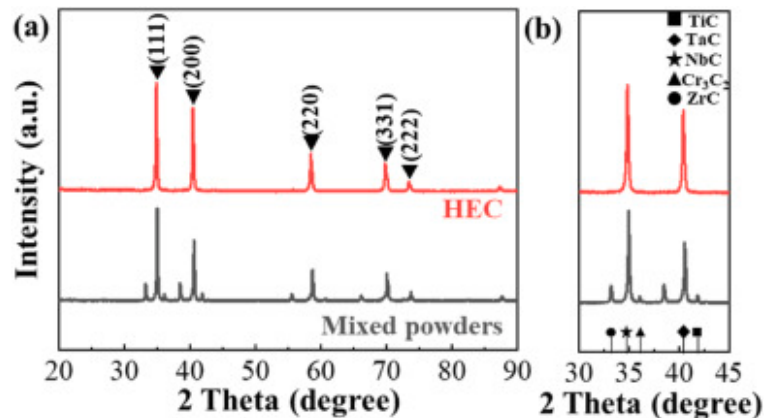
Brazing is an economical, straightforward, and highly effective joining technique widely used for joining both homogeneous and heterogeneous materials, particularly ceramics [[16], [17], [18], [19], [20], [21]]. Song et al. [22] used GNPs reinforced AgCuTi composite filler to join SiC ceramics. During the brazing process, it was observed that the graphene nanoparticles (GNPs) reacted with Ti, leading to the formation of TiC particles within the braze seam. This reaction successfully refined the microstructure of the brazed joint, indicating that the use of GNPs as a reinforcement was more effective in enhancing the mechanical properties of the brazed joint compared to using single filler materials. However, the operating temperature of AgCuTi solder is comparatively low. This limitation fails to meet the application requirements for ceramic joints in high-temperature environments. Sun et al. [23] employed a TiNiNb brazing filler metal to join (HfTaZrNbTi)C₅ high-entropy ceramics. The brazed joint primarily consisted of a TiNi+Nb eutectic structure, with (Ti,Nb)C and (Ti,Nb)₂Ni reaction layers forming on the ceramic surface, resulting in a strong bond. However, after 8 h of oxidation, excessive oxidation of the ceramic occurred, leading to a reduction in joint strength to 22 MPa. Currently, the joining mechanism of nickel-based brazing filler metals for welding high-entropy ceramics is not yet clear. There is an urgent need to find a high-temperature brazing filler metal that can be widely used in the brazing of high-entropy ceramics.

In this work, HEC/HEC joints were fabricated using a Ni-based brazing filler metal. The microstructure and interfacial evolution of the resulting joints were systematically investigated. The results indicate that the formation of the HEC/HEC joint primarily depends on Ni atom diffusion and the development of a Cr-C interfacial reaction layer. In addition, the change of joint strength with brazing temperature was also analyzed. This study aims to provide a feasible approach for achieving reliable HEC joints. The bonding mechanism of nickel-based brazing filler metal (TiZrNbTaCr)C ceramics was revealed to be stable bonding achieved through the diffusion of Ni into the HEC ceramic, the reaction of Cr with the ceramic, and the diffusion of Zr, Ti, and Nb elements from the ceramic into the brazing seam.

2. Materials and method

2.1. Experiments

The high-entropy carbide (TiZrTaNbCr)C used in this study was synthesized by hot pressing a mixed powder produced through ball milling, followed by treatment in a spark plasma sintering (SPS) device [24]. TiC, Cr₃C₂, ZrC, NbC and TaC powders with equimolar ratios of metal atoms were weighed. Since the molar ratio of C element to metal atoms in Cr₃C₂ is not 1:1, graphite powder was added to the mixed powder to make the molar ratio of metal atoms Ti, Cr, Zr, Nb, Ta and C elements in the mixed powder 1:1:1:1:1:5. Then, mechanical ball milling was performed at a speed of 300rpm for 24h to fully mix the powder. The ceramic powder designated for sintering was placed into a graphite mold, which was then positioned in a hot press furnace. The temperature was raised to 1750–1900 °C at a heating rate of 100 °C/min in an argon atmosphere, followed by sintering under a pressure of 50MPa for 10min. Subsequently, the high-entropy carbide ceramic block was cooled to room temperature inside the furnace. The phase structure illustrated in Fig. 1(a and b) indicates that the HEC ceramic obtained after spark plasma sintering exhibits a single rock salt phase structure, with no diffraction peaks observed for the five metal carbides, their solid solutions, or metal oxides.



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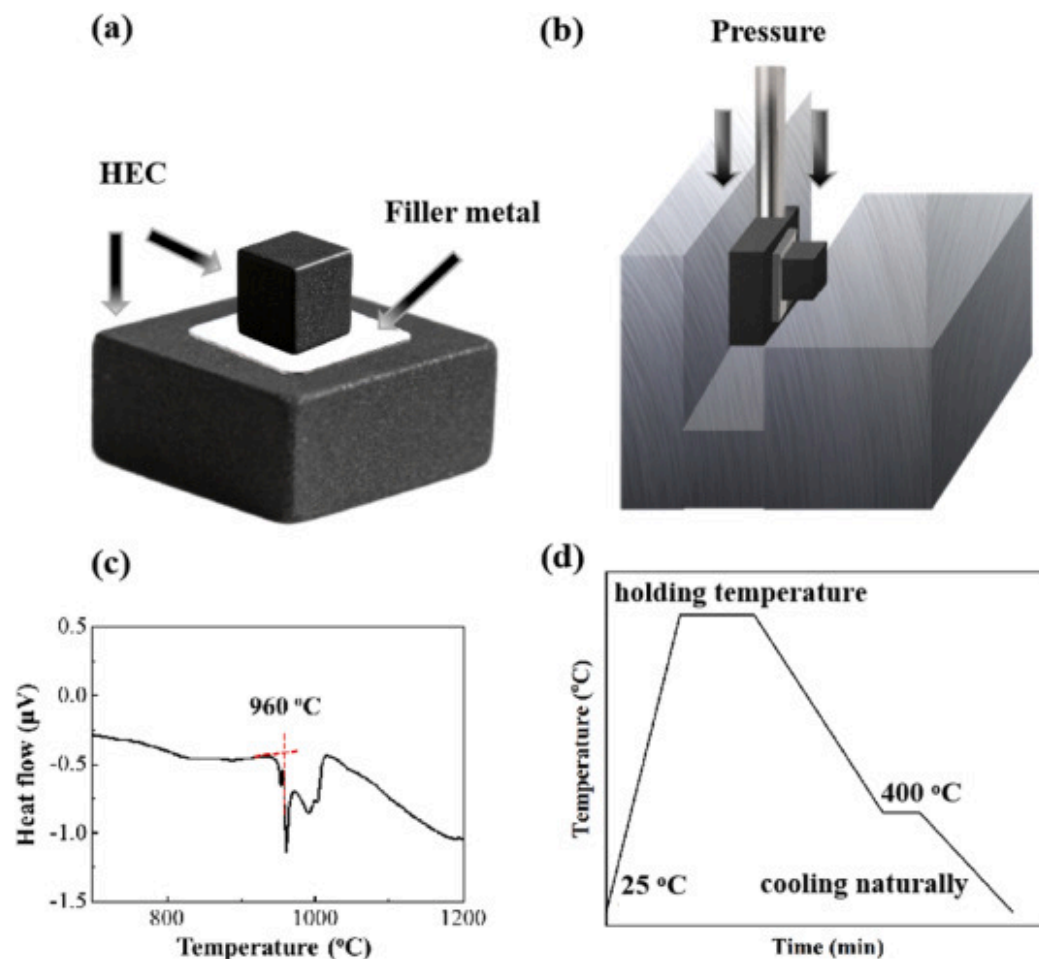
Fig. 1. The XRD patterns of HEC before and after sintering. (a) 20–90°, (b) 30–45°.

Ceramic specimens with dimensions of 5×5×5mm and 10×10×5mm were fabricated using a wire-cut electrical discharge machine. The HEC specimens were manually ground to a finish equivalent to 2000 grit using a diamond grinding wheel. In this study, commercially available BNi-2 filler was utilized. The filler thickness is 200μm, and the elemental contents are shown in

Table 1. The HEC specimens and BNi-2 filler were assembled as illustrated in [Fig. 2 \(a\)](#). The differential scanning calorimetry (Netzsch STA449F3 DSC) curve of the BNi-2 filler is presented in [Fig. 2 \(c\)](#). The BNi-2 filler exhibits a distinct endothermic peak with a melting point of approximately 960°C.

Table 1. Element composition of the brazing filler (at.%).

	Cr	Fe	Ni	Si	B
BNi2	6.77	3.47	82.55	4.16	3.05



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Fig. 2. (a) a schematic diagram of the brazing assembly, (b) hear strength test, (c) DSC result of BNi-2, (d) The process curve of brazing process.

After assembly, the samples were brazed in a vacuum furnace with the vacuum pressure maintained below 1×10^{-3} Pa. In the brazing experiment, to prevent misalignment of the joint, a graphite block was placed on top of the sample as a pressure block during welding. The thermal cycle profile of the brazing process is illustrated in Fig. 2 (d), with a holding temperature range of 1050–1200 °C and a holding time of 10 min.

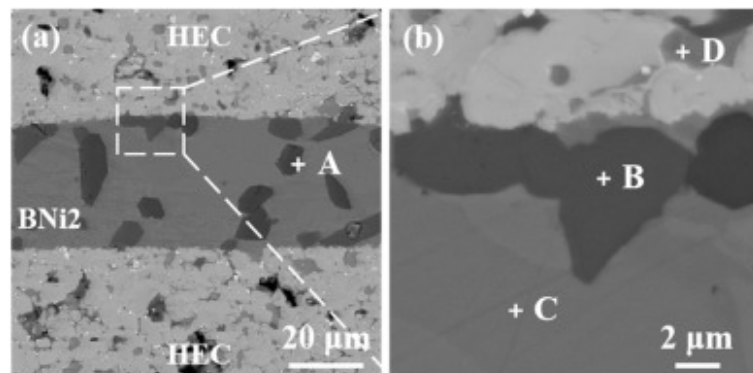
2.2. Materials characterization

The interface microstructure of HEC brazed joints was observed using backscattered electron (BSE) mode of scanning electron microscopy (Zeiss GeminiSEM 500, SEM). The chemical composition of the reaction products was analyzed using energy dispersive spectroscopy (Oxford Xplore30, EDS). X-ray diffraction (Bruker D8 ADVANCE, XRD) patterns of the brazed phase within the joints were collected to characterize their structures. Transmission electron microscopy (FEITEM Talos F200X TEM) was employed to observe and identify the reaction phases formed near the HEC matrix. TEM samples of the brazed joints were prepared using a focused ion beam (FIB). The schematic diagram of the shear strength testing device is illustrated in [Fig. 2 \(b\)](#). The shear strength of the brazed joints was evaluated at room temperature using a universal electronic testing machine. The loading rate was maintained at 0.5 mm/min [25]. At least three joints were tested, and the average value was used for analysis. SEM analysis was conducted to investigate the fracture morphology and microstructure.

3. Results

3.1. The typical microstructure of HEC/HEC brazed joint

The HEC/HEC joint, utilizing BNi-2 filler, was brazed at 1100°C for 10 min, and the typical microstructure of the HEC/HEC joint is shown in [Fig. 3](#). The brazing seam comprises a dark gray area corresponding to point A, a gray area corresponding to point B, and a light gray area corresponding to point C. In order to compare and analyze the composition of different phases in the joint, the EDS results for these points are presented in [Table 2](#). Zone A was primarily composed of 23.77% Ni and 71.77% Cr. It can be inferred that zone A corresponds to a Cr₇Ni₃ compound. According to the EDS results, it can be inferred that zone B and zone C correspond to the Cr-C reaction layer and Ni(s,s), respectively. Meanwhile, zone D, which contains Ni(s,s), is situated on the side of the HEC. The Cr and Ni elements play a crucial role in the interfacial connection.



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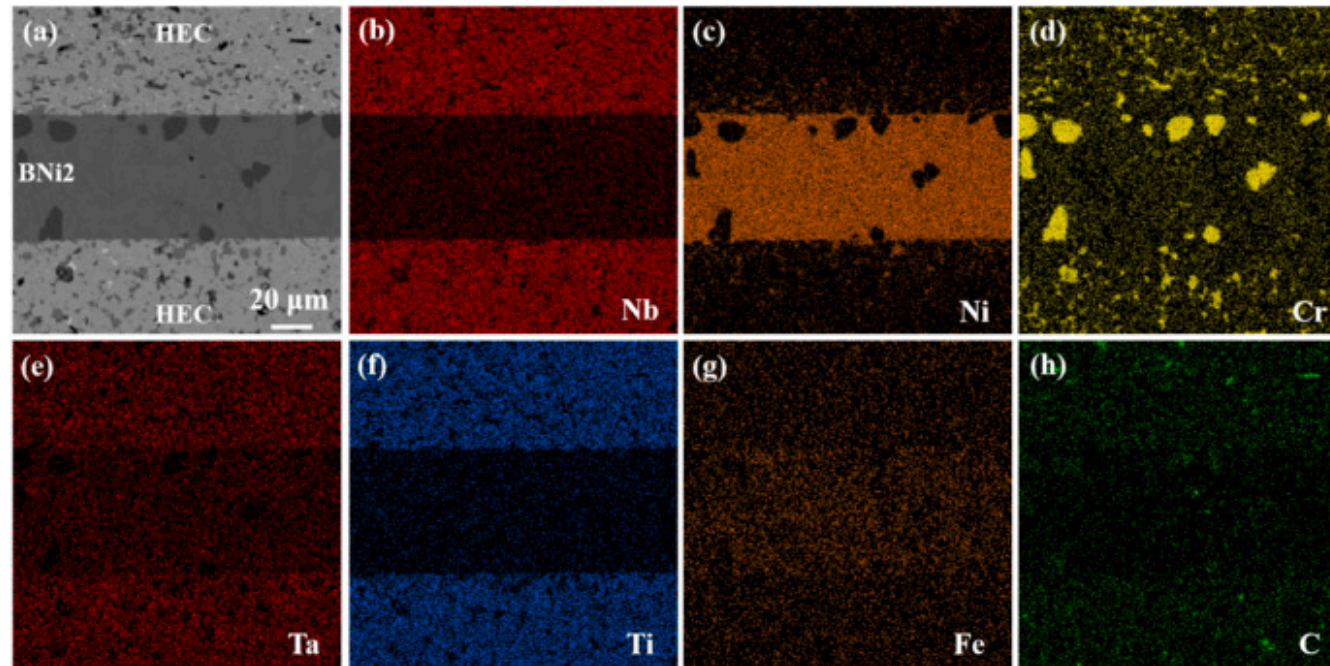
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Fig. 3. (a) The microstructure of HEC/HEC joint, (b) the enlarged image of interfacial reaction layer.

Table 2. Element composition of the marked points in Fig. 3 (at.%).

	Ni	Cr	Ti	Zr	Nb	Ta	C
A	23.77	71.77	0.12	0.38	1.35	2.61	–
B	0.68	42.26	0.09	0.10	1.35	0.15	55.37
C	91.14	3.01	0.10	0.47	4.03	0.92	0.33
D	91.31	1.24	0.98	0.23	1.43	0.83	3.98

To observe the overall elemental distribution in the weld, EDS surface scanning analysis was conducted on the joint. The results are presented in Fig. 4. As shown, the brazing seam is primarily composed of Cr, Fe, and Ni, which are all components of the BNi-2 brazing filler metal. Nb, Ti, and Ta, elements unique to HEC ceramics, are not found to be concentrated within the brazing seam. Furthermore, Ni clearly diffuses into the HEC ceramic and is concentrated within it, with a diffusion depth of approximately 20μm. Despite the extensive diffusion of Ni into the HEC ceramic, the ceramic maintains its integrity and is not dissolved by the brazing filler metal.



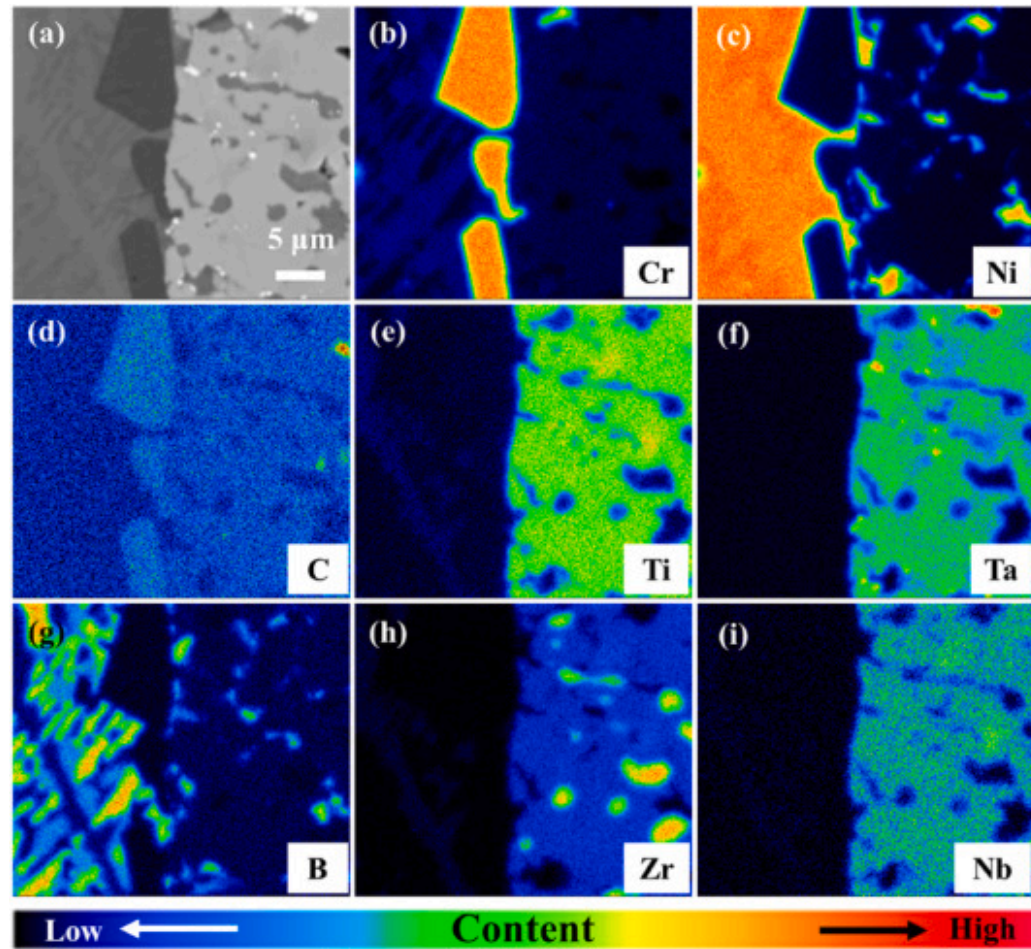
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Fig. 4. Element distribution at the interface of a HEC ceramic joint brazed with BNi-2 filler metal (a) Joint microstructure, (b) Nb, (c) Ni, (d) Cr, (e) Ta, (f) Ti, (g) Fe and (h) C.

To further investigate the reaction phase generated at the interface between the HEC and the central brazing seam, as well as the diffusion of Ni elements on the ceramic side, EPMA (Electron Probe Micro-Analysis) surface analysis was performed on the ceramic side area using an electron probe. The results are shown in Fig. 5. As illustrated in the figure, a significant aggregation of Cr elements is observed at the HEC interface. The gray reaction layer is primarily composed of Cr and C elements, indicating that Cr reacts with C to form a Cr-C compound reaction layer. As shown in Fig. 5(c–g), the Ni and B elements diffuse into the HEC, forming a clear segregation and distributing within the HEC [26,27]. It is speculated that during the brazing process, the liquid metal filler diffuses into the HEC along the grain boundaries, forming the Ni(s,s) phase. Therefore, the effective connection between the high-temperature brazing filler and HEC is attributed to the formation of the Cr-C reaction layer and the diffusion of the Ni element. There are various types of Cr-C compounds, including Cr₂₃C₆, Cr₇C₃ and Cr₃C₂. At 1400K, Cr reacts with C to form

Cr₂₃C₆, Cr₇C₃ and Cr₃C₂ in the following equation:



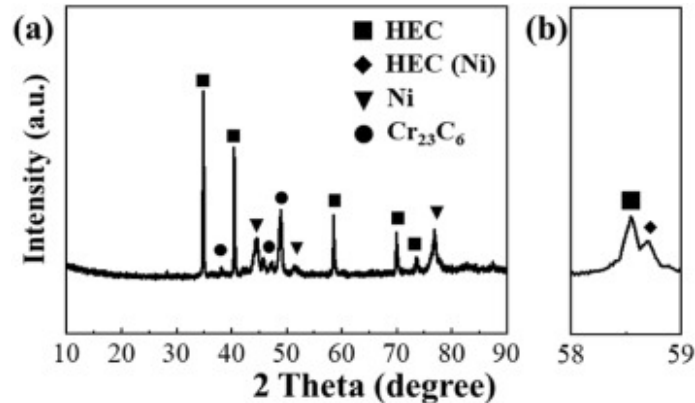
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Fig. 5. Element distribution at the ceramic-side interface of a HEC ceramic joint brazed with BNi-2 filler metal (a) Microstructure of the ceramic side, (b) Cr, (c) Ni, (d) C, (e) Ti, (f) Ta, (g) B, (h) Zr and (i) Nb.

Based on the Gibbs free energy of the formation of Cr-C compounds, it can be seen that the formation of Cr₂₃C₆ is most likely at 1400K (about 1127°C), Therefore, it is speculated that the phase generated at the interface between the ceramic and the brazed seam is Cr₂₃C₆ phase [28,29].

To further analyze the phase composition, we employed X-ray diffraction (XRD) analysis. Before the XRD test, the brazed joint was polished at the interface, and the weld was polished towards the HEC ceramic side to enhance the reaction. The sample was characterized using XRD, as shown in Fig. 6. As illustrated in Fig. 6(a), the reaction phase at the interface between HEC ceramics and the brazing joints is Cr₂₃C₆, which aligns with thermodynamic predictions; the connecting interface primarily consists of Cr₂₃C₆ and Ni(s,s). According to the amplified results of the XRD pattern in Fig. 6, within the 58° to 59° range, a small peak is observed on the right side of the diffraction peak of the originally single-phase HEC ceramics, which is attributed to the shift of the diffraction peak. This phenomenon occurs because the Ni element diffuses into the HEC ceramics during the brazing process and randomly occupies cation positions in the lattice of the HEC ceramics.

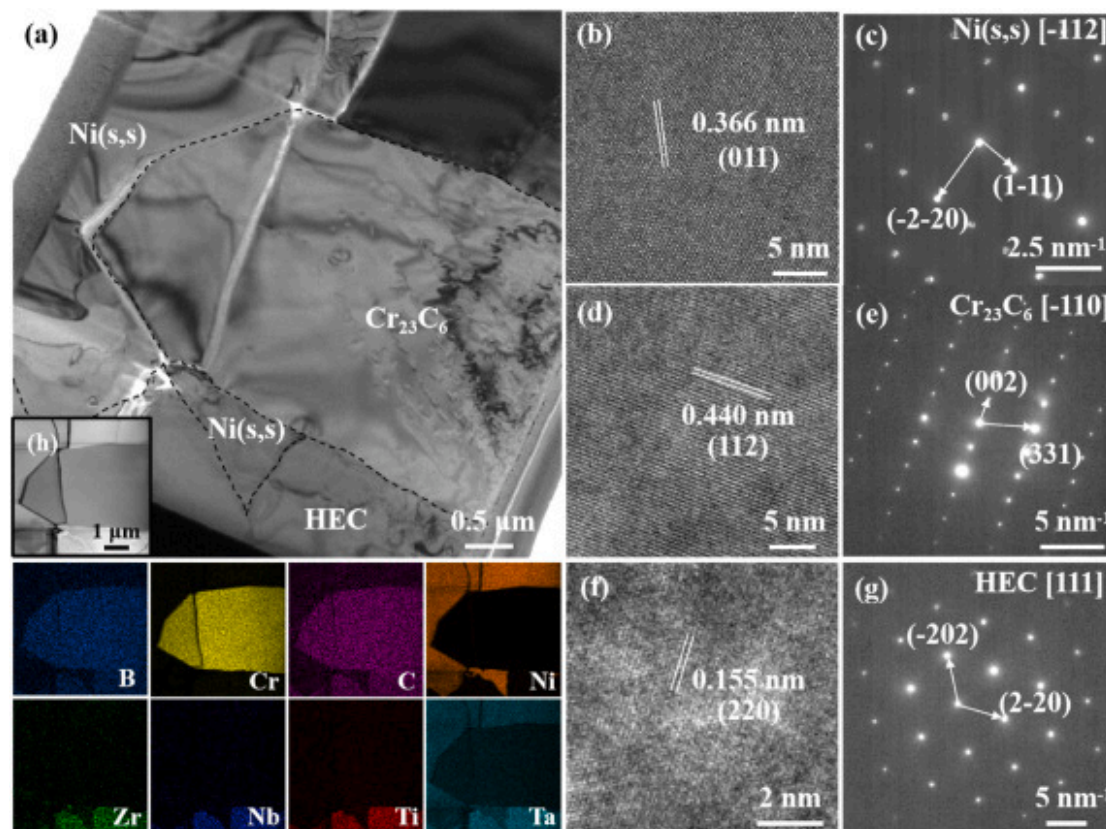


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Fig. 6. The XRD spectrum of the reaction phase on the side interface of HEC ceramics (a) 10°–90°; (b) 58°–59°.

Fig. 7 presents the transmission electron microscopy (TEM) analysis results of the thin reaction layer formed near the HEC substrate. In the bright-field image presented in Fig. 7(a), the Ni(s,s), Cr₂₃C₆, and HEC phases were identified. Inverse Fourier transform (IFFT) images and selected area electron diffraction (SAED) results for these three phases are illustrated in Fig. 7(b–g), respectively. These results confirm the existence of Ni(s,s), Cr₂₃C₆, and HEC reaction phases in the brazed joint. The elemental distribution illustrated in Fig. 7(h) indicates that the HEC phase is primarily composed of Zr, Nb, Ti, Ta, and C. Chemical reactions occurring at the interface between the Ni-based brazing filler metal and the HEC substrate during the entire brazing process resulted in the observation of Cr₂₃C₆ and Ni(s,s) phases in the reaction layer. The diffraction spots are located along the $[-1\ 1\ 0]$ zone axis of the Cr₂₃C₆ phase and the $[-1\ 1\ 2]$ zone axis of the Ni(s,s) phase, respectively. This finding indicates that during the brazing process, Ta and Ni atoms diffuse and combine to form an effective bond, while the formation of the Cr₂₃C₆ secondary phase strengthens the interfacial connection.



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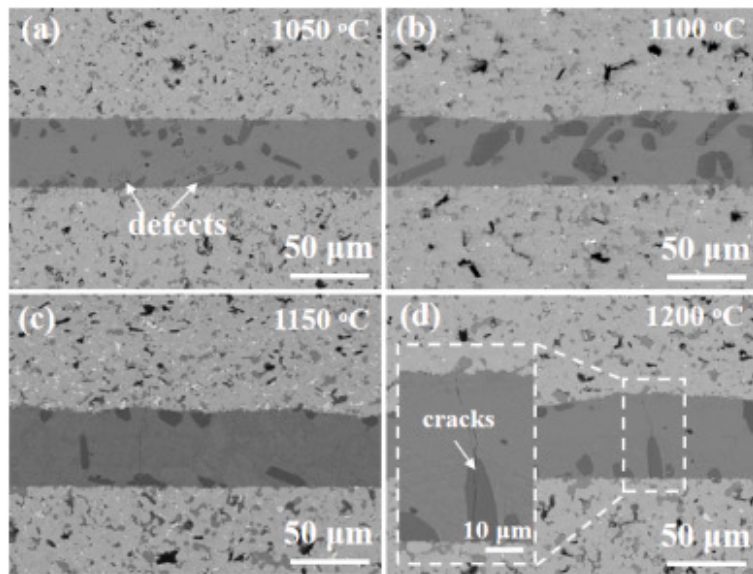
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Fig. 7. TEM characterization of HEC/BNi2/HEC joint (a)The bright field TEM of the interface structure; IFFT and SAED patterns of (b-c)Ni(s,s) $[-112]$ (d-e)Cr₂₃C₆ $[-110]$ (f-g) HEC $[111]$; (h) Elemental surface distribution.

In summary, the BNi-2 brazing filler metal successfully joined HEC ceramics at 1100°C for 10min. A small portion of the Cr in the filler metal accumulated in the brazed joint, forming a Cr₇Ni₃ phase with Ni, while a significant portion diffused into the ceramic side, resulting in the formation of a discontinuous Cr₂₃C₆ phase [30,31]. Ni in the filler metal diffused through the ceramic grain boundaries into the interior of the ceramic, ultimately forming Ni(s,s) around the HEC grains. Ni also diffused intragranularly into the HEC grains, randomly occupying cationic positions. Therefore, the stable connection achieved by BNi-2 brazing of HEC ceramics is primarily facilitated by the diffusion of Ni into the HEC ceramics, the reaction of Cr with the ceramics, and the diffusion of Zr, Ti, and Nb from the ceramics into the brazed joint.

3.2. The influence of brazing temperature on the microstructure and shear strength of HEC/HEC brazed joint

This section investigates the effect of brazing temperature on the microstructure and mechanical properties of HEC/HEC ceramic joints brazed with BNi-2 brazing filler metal, examining brazing temperatures of 1050°C, 1100°C, 1150°C, and 1200°C, with a fixed brazing time of 10min. The influence of brazing temperature on the joint was evaluated by analyzing changes in joint micromorphology, shear strength, and shear fracture micromorphology. Fig. 8(a–d) illustrates the micromorphology of the joint interface obtained at various temperature parameters. It can be observed that the brazing temperature did not significantly alter the brazing seam width, which remained approximately 40μm across all four temperature parameters. At lower temperatures, a substantial dark gray phase exists within the brazing joint, primarily located within the joint itself, with a smaller portion forming on the ceramic side. As discussed in the previous section, the dark gray phase within the joint is predominantly Cr₇Ni₃, while that on the ceramic surface is primarily Cr₂₃C₆. As the brazing temperature rises, the smaller Cr₇Ni₃ phase dispersed within the joint gradually decreases, while the Cr₂₃C₆ phase at the ceramic-joint interface increases. At a low brazing temperature (1050°C), some defects appear in the joint. This is due to the poor fluidity of the BNi-2 filler metal at low temperatures. As the temperature increases, these defects disappear. However, at 1200°C, microcracks develop within the joint due to the significant difference in thermal expansion coefficients between the HEC ceramic ($7.40 \times 10^{-6} \text{K}^{-1}$) and the BNi-2 filler metal ($13.0 \times 10^{-6} \text{K}^{-1}$).



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Fig. 8. The microstructure of HEC/HEC joint brazed at (a) 1050°C, (b) 1100°C, (c) 1150°C, (d) 1200°C.

The above joint microstructure analysis indicated that Ni(s,s), Cr₂₃C₆, and Cr₇Ni₃ are the primary phases in HEC joints brazed with Ni-based filler metals. To further elucidate the brazed joint joining mechanism, the evolution of the Gibbs free energy for each phase's formation was analyzed. From a thermodynamic perspective, the formula for calculating the Gibbs free energy in an ideal solid solution is as follows:

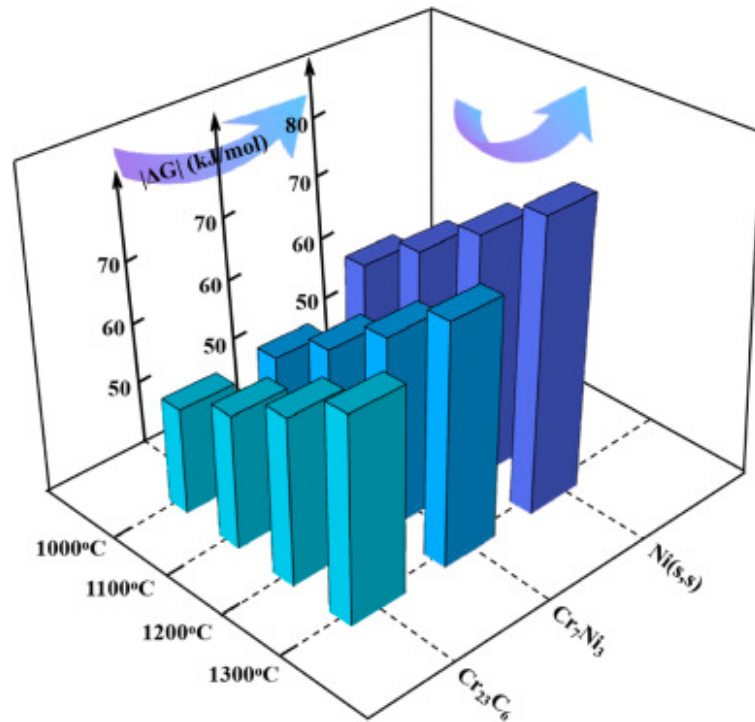
$$\Delta G_{mix} = \Delta H_{mix} - T_K \Delta S_{mix} \quad (4)$$

Where ΔH_{mix} represents the enthalpy of mixing, ΔS_{mix} represents the entropy of mixing, and T_K represents the absolute temperature. The Gibbs free energy formulas for the two compounds Cr₂₃C₆ and Cr₇Ni₃ generated in Fig. 9 are as follows:

$$\Delta G = \Delta G^\theta + RT \ln \left(\frac{p}{p^\theta} \right) \quad (5)$$

$$\Delta G^\theta = H_T - H_{298} - T\varphi'_T \quad (6)$$

Where ΔG^θ represents the standard Gibbs free energy (kJ/mol^{-1}), R represents the gas constant, p and p^θ represent the vacuum pressure and standard atmospheric pressure, respectively, T is the reaction temperature, and φ'_T is the Gibbs free energy function of the substance.



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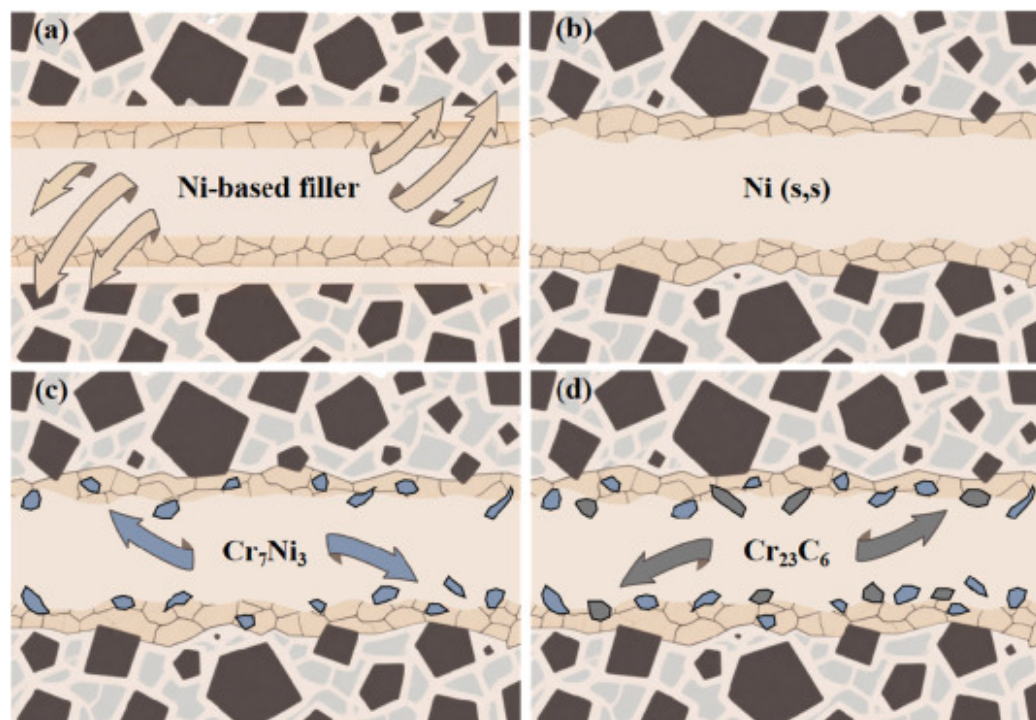
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Fig. 9. Molar Gibbs formation free energy of each phase in the brazed joint.

The Gibbs formation energy of each phase was calculated using JMatPro version 8.0. The absolute value of the Gibbs free energy, $|\Delta G|$, characterizes the magnitude of the reaction's driving force; a larger value indicates a stronger driving force. The brazing reaction is typically performed in a vacuum hot press furnace and can be considered a quasi-steady-state process. Despite the presence of nonequilibrium conditions, the Gibbs free energy remains an effective thermodynamic criterion, allowing the formation priority of each phase in the joint to be inferred, thereby revealing the joint joining mechanism. The results were shown in Fig. 9. The absolute value of the Gibbs free energy of formation for the Ni(s,s) phase is the largest, while that for Cr₂₃C₆

is the smallest. This indicates that the Ni(s,s) phase forms first in the weld, while Cr₇Ni₃ and Cr₂₃C₆ form later than the Ni(s,s) phase and are therefore dispersed within the Ni(s,s) phase. This is consistent with the brazing experimental results.

The brazing mechanism of HEC brazed with Ni-based filler metal is summarized in Fig. 10. The brazing metallurgical process is primarily divided into three stages. In the first stage, after the filler metal melts, reaction and diffusion occur between the liquid phase and the matrix. In particular, the Ni element in the filler metal diffuses into the ceramic through the grain boundaries and into the interior of the HEC grains through intragranular diffusion, randomly occupying cationic positions. In the second stage, due to differences in elemental affinity and metallurgical compatibility, some elements in the liquid phase segregate. A small amount of Cr in the filler metal aggregates in the braze joint and forms a Cr₇Ni₃ phase with the Ni. In the third stage, another portion of the Cr element that diffuses toward the ceramic side forms a discontinuous Cr₂₃C₆ phase on the ceramic side, distributed within the Ni(s,s). The two hard phases, Cr₇Ni₃ and Cr₂₃C₆, are evenly distributed within the soft matrix, providing a strengthening effect. The composition and structure of ceramics and metals differ significantly, and direct bonding can easily produce a brittle interface layer. However, the composition of the hard phase is similar to that of the ceramic parent material, resulting in good structural compatibility. This hard phase acts as a "stress transfer bridge," smoothly transferring external loads from the parent material to the weld, preventing weld fracture caused by stress concentration.



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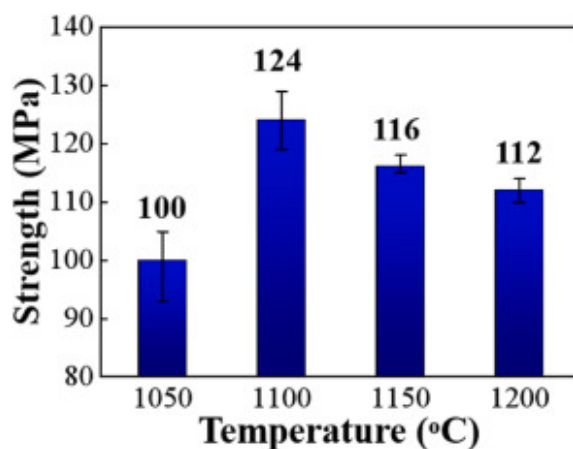
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Fig. 10. Schematic for the microstructural evolution of HEC brazed joint with Ni-based filler during the brazing process:(a) filler melting and elemental diffusion stage, (b) Ni(s,s) was generated during cooling, (c) Cr₇Ni₃ was generated during cooling, (d) Cr₂₃C₆ was generated during cooling.

In summary, in the joints brazed with HEC using Ni-based filler metal, the Ni(s,s) phase preferentially formed, and a continuous brittle phase interface reaction layer was absent on the HEC side, achieving a high-quality HEC connection.

Fig. 11 illustrates the shear strength of HEC/HEC ceramics at various brazing temperatures with a fixed brazing time of 10 min. As the temperature increases, the shear strength of the ceramic joint initially increases rapidly, then gradually decreases. At 1050°C, insufficient element diffusion within the joint, poor filler metal fluidity, and defects in the braze seam result in joint strength lower than at higher temperatures, reaching only 100 MPa. When the temperature rises to 1100°C, element diffusion becomes more complete, and the joint reaches a strength of 124 MPa. When the temperature is further increased to 1200°C, microcracks

form in the braze seam due to the difference in thermal expansion coefficients between the HEC ceramic and the BNi-2 filler metal, resulting in a further decrease in strength.

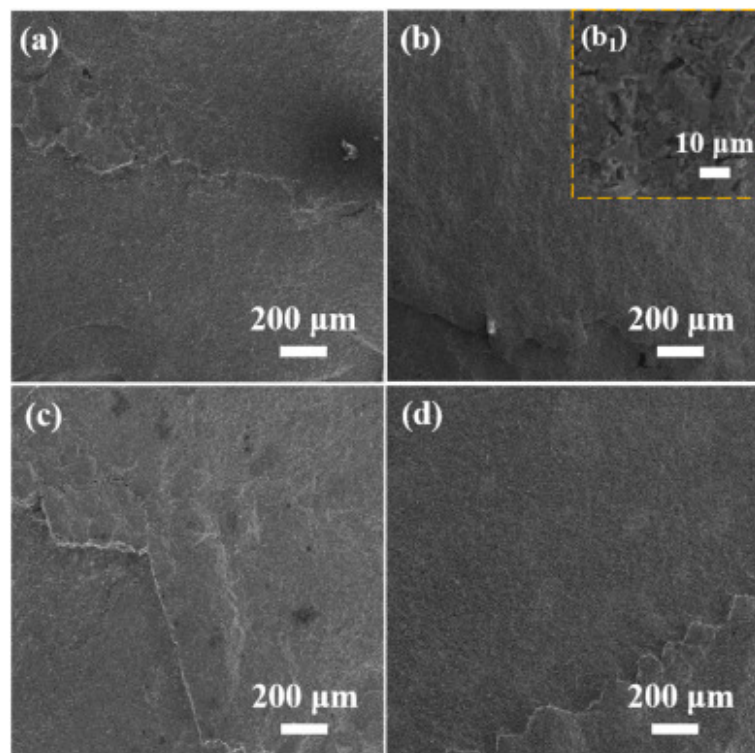


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Fig. 11. The shear strength of HEC/HEC joints brazed at different brazing temperature.

The fracture morphology of HEC/HEC joints at different temperatures is shown in Fig. 12. It can be observed from the figures that the overall fracture morphologies of the joints vary significantly with temperature. At a brazing temperature of 1050°C, the joint fractures at the ceramic near the brazed seam, resulting in a relatively smooth surface that exhibits a typical brittle fracture pattern. As the temperature rises to 1100°C, the joint fractures primarily on the ceramic side, indicating higher shear strength. With a further increase in temperature to 1150°C and 1200°C, the joint initially fractures at the ceramic near the brazed seam, then propagates through the brazed seam and continues near the seam on the opposite side. This phenomenon is attributed to the high residual stress in the parent material near the brazed seam, which makes this region a mechanically weak point and prone to initial fracture when pressure is applied.

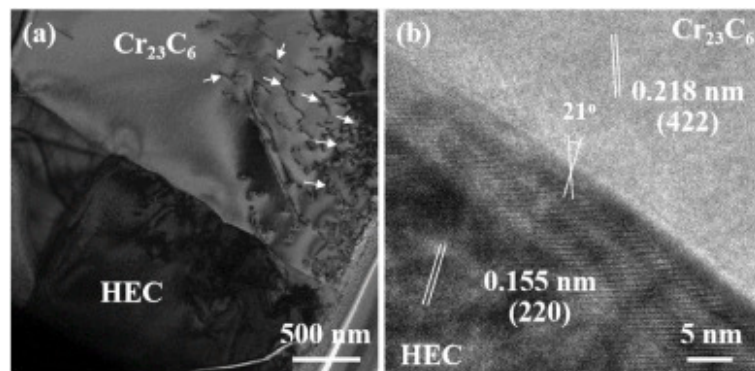


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Fig. 12. Fracture morphology of joints at different brazing temperatures under room temperature shear; (a) 1050°C, (b) 1100°C, (c) 1150°C, and (d) 1200°C.

In the bright-field TEM image of Fig. 13 (a), a large number of slip traces and deformation bands can be clearly observed in the Cr₂₃C₆ phase, indicating that this area has been subjected to large thermal stress during the cooling process and has achieved partial stress release through its own plastic behavior. In the bright-field TEM image of Fig. 13(a), numerous slip lines and deformation bands are clearly observed within the Cr₂₃C₆ phase, indicating that this region experienced significant thermal stress during cooling and achieved partial stress relief through its own plastic behavior. A clear interface between HEC and Cr₂₃C₆ can be observed in Fig. 13(b). Furthermore, the angle between the two phase planes is 21°, indicating a small lattice mismatch and low interfacial energy.



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Fig. 13. TEM analysis of HEC joint close to the HEC side (a) Morphologies of main phases (b)HRTEM image of the Cr₂₃C₆/HEC interface.

4. Conclusion

This study used brazing technology to successfully achieve reliable bonding of (TiZrTaNbCr)C ceramics using nickel-based brazing filler metals, demonstrating excellent mechanical properties. Based on this, the following conclusions were drawn:

- (1) BNi-2 brazing filler metal was used to braze (TiZrTaNbCr)C ceramics at 1050–1200°C. During the brazing process, the Cr element in the BNi-2 brazing filler metal only formed a discontinuous Cr₂₃C₆ phase on the HEC surface. The Ni element achieved good diffusion into the ceramic. It not only diffused from the grain boundary into the HEC ceramic to form Ni(s,s) in the ceramic, but also diffused into the HEC ceramic grains and randomly occupied the metal atomic positions. This is the main reason why the BNi-2 brazing filler metal can achieve HEC ceramic connection.
- (2) As the brazing temperature increases, the smaller Cr₇Ni₃ phases dispersed inside the brazing seam gradually decrease. At the same time, the Cr₂₃C₆ phase at the interface between the HEC and the brazing seam gradually increases and continues to grow with the increase in temperature. Among them, Cr₂₃C₆ exists as a second phase in the brazing seam, which plays a role in improving the joint strength. However, when the

temperature reaches 1200°C, due to the obvious difference in thermal expansion coefficients between the HEC ceramic and the BNi-2 brazing filler metal, microcracks are generated in the brazing seam.

- (3) Finally, the shear strength and fracture morphology of the (TiZrTaNbCr)C joints at different brazing temperatures were studied. The shear strength of the joint initially rose quickly before gradually declining as the temperature increased. The peak shear strength recorded for the joint was 124MPa at a brazing temperature of 1100°C for 10min. The fracture predominantly occurred on the ceramic side of the joint.

CRedit authorship contribution statement

Jinzheng Li: Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Pengguan Wang:** Software, Resources, Project administration, Conceptualization. **Pengcheng Wang:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Data curation. **Zhaoyi Pan:** Project administration, Methodology, Investigation, Funding acquisition. **Yaotian Yan:** Resources, Methodology, Funding acquisition, Data curation, Conceptualization. **Xiaoguo Song:** Writing – review & editing, Methodology, Investigation. **Haiyan Chen:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Formal analysis. **Wenya Li:** Visualization, Validation, Supervision, Software, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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